EFFECTS OF CAFFEINE ON BOTTOM-UP PERIPHYTON-GRAZER INTERACTIONS IN AQUATIC ECOSYSTEMS

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ABSTRACT

Kevin McCluney, Committee Chair

Caffeine is one of the most widely used pharmaceutical substances in the world and is consequently detected to some extent in many aquatic environments impacted by human wastewater. While a wide range of concentrations have been observed, varying considerably across time and space, and while direct effects have been noted on various animal taxa, relatively little work has been done on understanding the indirect impacts of caffeine on animals via effects on the base of the food web. In order to help address this gap, I conducted a study in which I grew periphyton in a controlled setting with varying levels of caffeine exposure and then performed a feeding trial on this periphyton using snails that had not been directly exposed to caffeine. My findings suggest that only high concentrations of caffeine (0.5 g/L but not 1 mg/L in my study) have a noticeable effect on the biomass of the periphyton community, causing it to decrease likely due to toxicity. However, even low concentrations of caffeine can result in significant increases to the nitrogen content of the periphyton, potentially due to the use of caffeine as a nitrogen source. While snails did not significantly change their feeding behavior or grazed area across most of the concentrations, snails feeding on the periphyton exposed to the highest concentration of caffeine were more mobile and less likely to remain on the periphyton tile. Whether this is caused by secondary exposure to caffeine remaining within the biofilm or due to other changes in the biofilm is uncertain. These results provide useful insights into the potential ecological impacts of caffeine as a contaminant, as well as representing a basic methodology for studying ecotoxicology from an indirect, bottom-up approach in isolation from direct effects.

In memory of my father and my grandmother,

whose love of learning and of nature continues to inspire me.

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INTRODUCTION

Anthropogenic contaminants have long been recognized as a potential threat to aquatic ecosystems. While many of these contaminants have been subject to extensive study (Mansouri et al., 2017; Perhar and Arhonditis, 2014), others have only recently received significant scientific attention (Sauvé, 2014). One category of such contaminants of emergent concern are pharmaceutically active compounds, such as those found in medications for humans and livestock, illicit drugs, and certain personal care products (Cizmas et al., 2015; Fontes et al., 2020; Montes-Grajales et al., 2017). Complex mixtures of these chemicals are released into aquatic ecosystems by human activity, with a major source being the effluent of wastewater treatment plants (WWTPs; Marshall & McCluney, 2021; Schoenfuss et al., 2016). Once introduced, both the contaminants themselves and their indirect effects can be propagated through the food web by a variety of pathways, including some that can cross the boundary between aquatic and terrestrial ecosystems (e.g., Walters et al., 2008)

Caffeine as a Contaminant of Emergent Concern

A particularly widespread pharmaceutically active contaminant is caffeine, an alkaloid stimulant and one of the most highly consumed psychoactive substances in the world. Plants such as coffee, tea, and cacao produce caffeine naturally as a secondary metabolite with documented anti-insect (Nathanson, 1984), anti-bacterial (Sledz et al., 2015), anti-fungal (Wang et al., 2016), and even allelopathic properties (Pham et al., 2019). However, these plants have a geographically limited range. Products made from these plants for human consumption, meanwhile, are much more widespread, to such a degree that the detection of caffeine in many areas can be considered an indicator of human wastewater contamination (Seiler et al., 1999; Kolpin et al., 2002; Peeler et al., 2006). Caffeine is broken down with relatively high efficiency

in the human body (Tang-Liu et al., 1983), in wastewater treatment plants (Bruton et al., 2010), and in the environment (Lam et al., 2004). Nevertheless, its high rate of consumption results in constant replenishment and a persistent presence in many aquatic ecosystems (Rodríguez-Gil et al., 2018; Quadra et al., 2020). Furthermore, especially high levels of caffeine in aquatic ecosystems may be indicative of contamination by untreated wastewater, such as from the septic tank leakage (Seiler et al., 1999), from overflowing sewage systems and WWTPs during periods of high rainfall or snowmelt (Buerge et al., 2003; Hillebrand et al., 2012), or from inadequate treatment practices (Li et al., 2020b).

Because it is primarily introduced in human wastewater, caffeine's ecological impact is most significant in systems downstream of WWTPs serving large human populations. In larger rivers and lakes, the volume of water will more effectively dilute the effluent and reduce the concentration of caffeine and other wastewater contaminants, though areas such as estuaries which lie downstream of multiple WWTPs may have somewhat higher concentrations for their size (Rodríguez-Gil et al., 2018). However, in smaller streams, effluent flow can make up a much greater proportion of the total flow, which can result in higher concentrations of chemicals introduced by effluent (Du et al., 2015; Brooks et al., 2006). Because small, low-order streams make up sizeable portions of the length of most river systems and form a significant interface between aquatic and terrestrial systems (Downing et al., 2012; Wohl, 2017), assessing the ecological impact of contaminants such as caffeine on such systems is of substantial importance.

Although the total ecotoxicological impact of caffeine requires further study, notable effects have been observed on various organisms, at both experimental and environmentally realistic concentrations (Korekar et al., 2020; Li et al., 2020a). At high concentrations (~5 mg/L to ~100 mg/L), caffeine has been observed to induce morphological abnormalities in embryonic

fish, as well as increases in anxiety-indicating behavior (Lee & Wang, 2015) and damage to lateral line hair cells (Rah et al., 2017). Caffeine has also been observed to reduce sleep and increase movement behavior in some invertebrate species, similar to its effects on mammals (Mustard, 2014). Oxidative stress and increased mortality were observed in polychaetes (Pires et al., 2016). Relatively little research has been done on the effects of caffeine on aquatic snails, but studies on terrestrial snails and slugs found that caffeine had toxic effects (Wszelaki et al., 2023) and could potentially act as an agricultural repellant (Hollingsworth et al., 2003; Horgan et al., 2023). On a larger ecosystem scale, experimental introduction of caffeine to a stream led to a decrease in insect emergence rates nearby (Marshall, 2019), which could have repercussions for their terrestrial predators. Overall, the broad variety of organisms that may be affected and the widespread consumption and environmental presence of caffeine warrant further study of its effects on organisms and ecological interactions at a wide range of concentrations.

Caffeine-Periphyton-Grazer Interactions in Stream Ecosystems

Within the broader context of stream ecosystems, caffeine may also have an effect lower on the food web. Specifically, it may have an impact on the complex microbial communities found in periphyton biofilms. These complex communities of algae, bacteria, fungi, and other organisms exist in a mucilage biofilm on surfaces submerged in water and can contribute significant amounts of primary productivity, especially in less forested landscapes (Hagen et al., 2010). This production enables them to serve as a food source for a diverse guild of macroscopic grazers. These grazers are in turn consumed by fish and other predators, some of which may be from outside the stream itself. For example, terrestrial species such as raccoons may prey on grazing aquatic snails, and some grazers such as emergent insects and amphibians may leave the stream at some point in their life history (Baxter et al., 2005). As such, the effects of

contaminants such as caffeine on these species and their interactions are ecologically relevant both within streams and in neighboring terrestrial ecosystems.

One way caffeine may interact with grazers and periphyton is the potential for biomagnification. A few recent studies have found evidence that caffeine can accumulate within periphyton (Burket et al., 2020; de Sousa et al., 2021). If this occurs, it may expose grazing organisms to higher concentrations of caffeine than are present in the water itself. However, since caffeine can be metabolized by a variety of bacteria (Summers et al., 2015), it may not be retained in the biofilm in a manner conducive to the sorts of biomagnification seen in more persistent organic contaminants such as PFAS (Ecke et al., 2024). Indeed, some studies have suggested that feeding exposure for snails is less important than direct exposure in water (Du et al., 2015). One recent study (Yang et al., 2020) found that caffeine tended towards trophic dilution rather than trophic magnification, while another (Sokołowski et al., 2024) found more neutral results, with no clear indication of either trophic magnification or trophic dilution.

Beyond acting as a vector for caffeine exposure, direct effects of caffeine on periphyton may have indirect, bottom-up effects on other taxa. If caffeine has a negative effect on the total biomass of periphyton, as suggested by some studies (e.g., Rosi-Marshall et al., 2013), this will limit the amount of food available for grazers. This may occur because of toxicity, or it may occur because of shifts within the periphyton community (Lawrence et al., 2012), possibly affected by the availability of light and other limiting factors such as phosphorus availability (Marshall, 2019). Alternatively, some research suggests that periphyton biomass may increase with the presence of caffeine, possibly because it can act as a nitrogen source for some taxa, such as Betaproteobacteria (Lawrence et al., 2005) or various Nucletmycea such as fungi (Marshall, 2019). Regardless of effects on the total biomass, caffeine may also influence the nutrient

concentrations within biofilm, potentially increasing its nitrogen content (Marshall et al., 2022). It is also possible that these effects may vary with the specific concentration of caffeine present (de Sousa et al., 2021).

Both biomagnification and bottom-up food web effects are fairly well studied as general concepts, although the specific application of these concepts to many individual chemical contaminants remains understudied (Bernhardt et al., 2017). Furthermore, the total impacts may be more complex. For example, shifts in the periphyton community (e.g., Lawrence et al., 2012) may affect the efficiency of grazing organisms, since some taxa such as diatoms may be easier for grazers to process than others (Brönmark, 1989). Changes in the nutrient content of periphyton may alter nutrient limitations for higher trophic levels, possibly inducing compensatory feeding behavior to accommodate reductions in food quality (Iannino et al., 2019). Finally, if caffeine does accumulate within the biofilm of periphyton, high concentrations may prompt avoidance behavior and lower the rate of feeding (Hollingsworth et al., 2003).

Overall, there are several potential pathways for interactions between caffeine, periphyton, and grazers (Figure 1). For my thesis, I have focused on the bottom-up pathways, investigating the impact of exposure to different caffeine concentrations on periphyton growth and nutrient content, as well as the indirect effects on snail grazing behavior when feeding on this periphyton without direct exposure to caffeine in the water.

If caffeine is predominantly acting as a toxin to periphyton at a given concentration, I would expect reductions in periphyton biomass. If instead it is predominantly acting as a nutrient source, I would expect increases in periphyton biomass if nitrogen is a limiting nutrient. In this case, I would also expect alterations to the nutrient content of the periphyton if it has a flexible stoichiometry or undergoes a shift in community composition. An increase in grazing quantity

during feeding might indicate a shift away from the snail's preferred nutrient content (requiring compensatory feeding), while a decrease in grazing quantity might indicate a shift towards that preference (causing faster satiation). Alternatively, increased grazing quantity could be attributed to a behavioral effect of lingering caffeine within the biofilm (which I would expect to be accompanied by higher overall movement), while a decrease could also be attributed to an avoidance behavior induced by caffeine toxicity (which I would expect to be accompanied by changes in the snails' location preferences during the feeding trial).

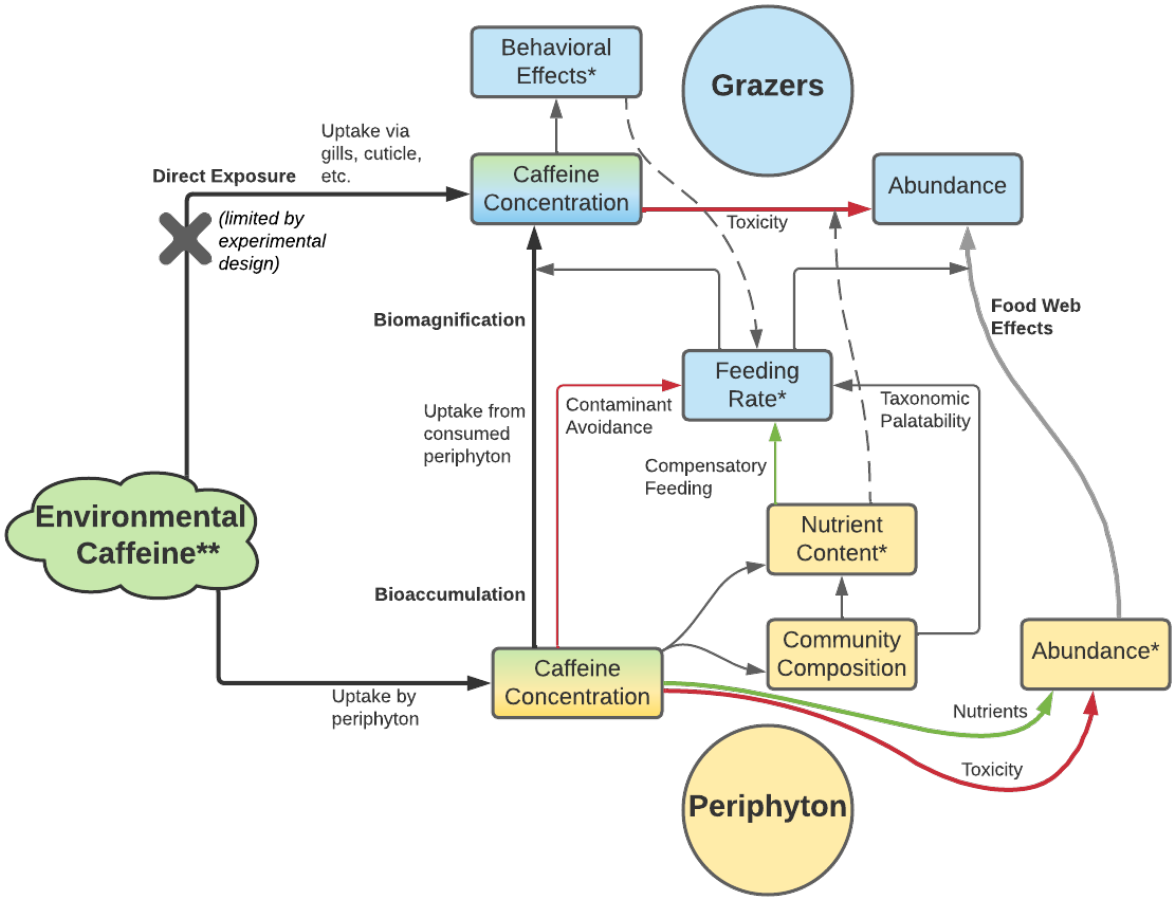

Figure 1: Potential caffeine-periphyton-grazer interactions (** indicates the independent variable, whereas * indicates dependent variables to be measured). Black arrows indicate the flow of caffeine, red arrows indicate negative effects, green arrows indicate positive effects, and grey arrows indicate mixed or uncertain effects.

METHODS

In order to test my hypothesized effects of caffeine on periphyton, I grew periphyton in a laboratory setting under different exposures to caffeine. I then conducted a feeding trial with snails to assess the indirect effects of this caffeine exposure on periphyton consumers (Figure 2).

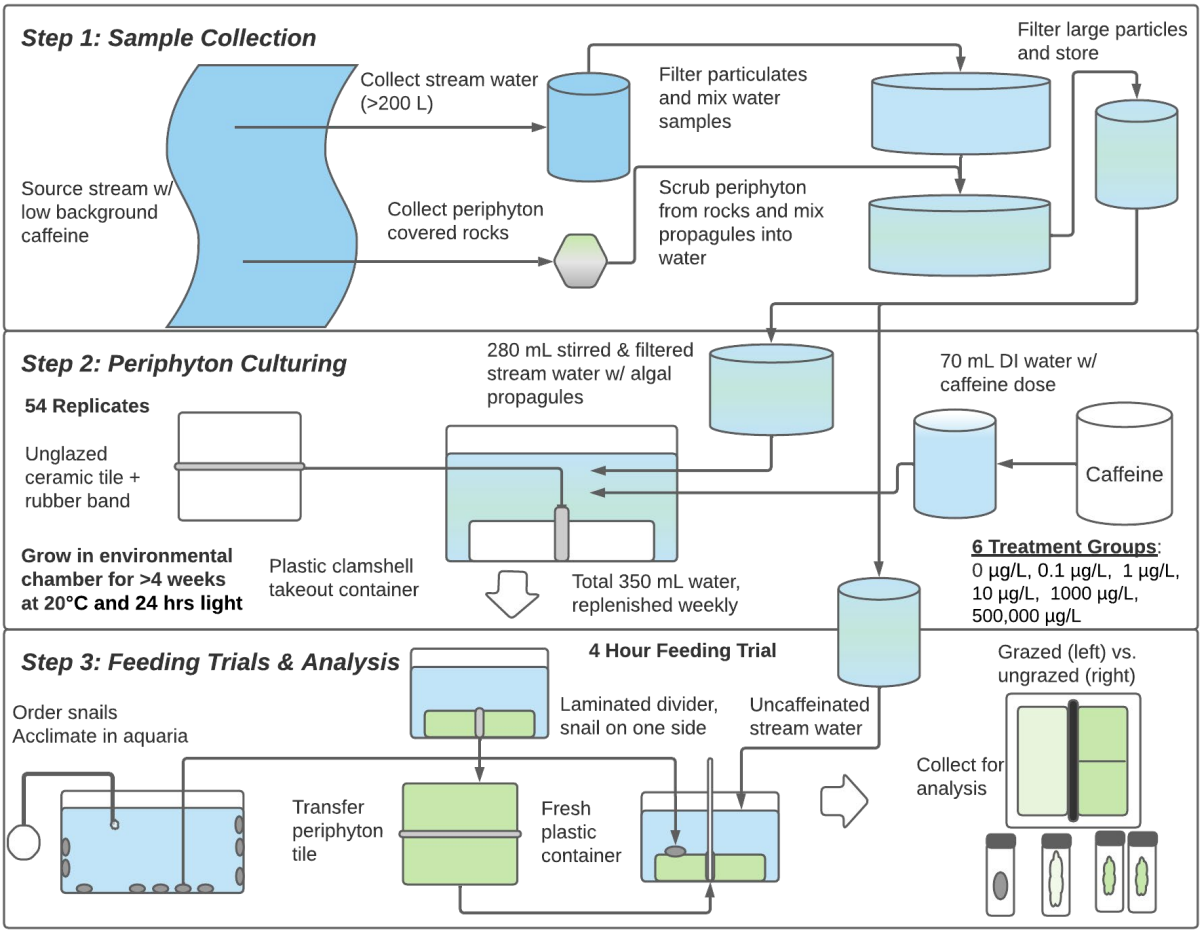

Figure 2: Graphical overview of experimental protocols. Step 1 describes sample collection and processing. Step 2 describes periphyton growth setup. Step 3 describes snail feeding trial.

Water, Periphyton, and Snails

In order to ensure appropriate nutrients for the growth of the periphyton, natural stream water was procured (Figure 2, Step 1). The source chosen was Sycamore Creek, a tributary of the Sandusky River in Wyandot County, Ohio (Figure 3; Appendix Figure 17). Prior work had

established this site as one with low background levels of caffeine (Marshall et al., 2022), which ensured that the experimental addition of caffeine would be the predominant source. A surplus of stream water was collected for the experiment and stored in buckets in an unlit climate-controlled chamber at 4°C when not in active use. In addition to the water, approximately 20 sizeable rocks coated in periphyton were collected from the stream bed and subsequently stored in the water sample.

For the feeding trial, freshwater snails (Nerites sp.) were ordered from Carolina Biological Supply Company. These were housed by the BGSU Marine Lab until needed.

Figure 3: Sycamore Creek, a tributary of the Sandusky River, Ohio, near where water and rocks were acquired.

Periphyton Growth

All stream water samples were mixed together in a large container to homogenize them. Periphyton from all collected rocks was scrubbed into this collected water to suspend an

abundance of periphyton propagules. After the water and periphyton had been thoroughly mixed, the water was returned to storage buckets through a Hubbard Scientific Co #60 fine mesh (250 micron) sieve to remove large particles and exclude unwanted invertebrate animals from the experiment. Water samples were then returned to cold storage (Figure 2, Step 1).

Growth containers were created for the experiment using plastic clamshell takeout containers cut in half (interior dimensions 12.7 cm x 12.7 cm, and at least 3 cm deep). To provide a substrate for periphyton growth, unglazed ceramic tiles (10 cm x 10 cm x 0.6 cm) were added to each container. Each tile was bisected with a thick rubber band (0.6cm wide), dividing it into two portions for the later feeding trial. A handle made from fishing line was attached to this rubber band in order to limit direct contact with the tile during upkeep and minimize disturbance to the biofilm.

A total of six treatment groups were designated with nine replicates each, for a total of 54 individual tiles (Figure 2, Step 2). The control group received no added caffeine. The remaining five treatment groups spanned a wide range of concentrations comparable those observed in various literature studies, including 0.1 µg/L (approx. median for NW OH streams from Wu et al., 2009), 1 µg/L (approx. median for European streams from Zhou et al., 2019), 10 µg/L (overlap of WWTP influent/effluent values from Buerge et al., 2003), 1 mg/L (based on a particularly high environmental value documented near a coffee bean processing plant in Costa Rica by Spongberg et al., 2011), and 0.5 g/L (an extreme value similar to the concentration of a cup of coffee, from Burdan, 2015).

Replicates were divided into three batches of 18, with 3 replicates of each of the 6 treatment groups within each batch (9 total replicates across batches). These batches were grown

concurrently but offset in time by about 6 days to spread out the upkeep and keep the simultaneous feeding trials to a workable number.

At the start of the 8-week growth period and on a weekly basis for the duration, water in each container was replaced. The slight variation in concentration created by this weekly replacement may mimic pulsed patterns of exposure to certain pharmaceuticals observed in nature as a result of temporal dynamics in human pharmaceutical usage (McCluney Lab, Unpublished Data). The periphyton tile was lifted out of its container by its handle, the old water was dumped out of the container, new water was added with a known concentration of caffeine, and finally the periphyton tile was returned to the container. This process generally took under 30 seconds, ensuring minimal drying to the tile.

A total of 350 mL of new water was added to each replicate at each replacement. 280 mL of this was stream water from Sycamore Creek, removed from cold storage and with periphyton propagules re-suspended by stirring. The remaining 70 mL was DI water in which an amount of caffeine equal to 5x the target concentration for the replicate was dissolved ahead of time (to meet the target in the final 350 mL). Lower concentrations of this 70 mL dose were created by diluting excess portions of higher concentrations. The control group instead received 70 mL of DI water with no added caffeine.

Once the replicates were prepared, they were placed into a Conviron reach-in environmental chamber (Figure 4) with Halco ProLume Eco-Shield fluorescent lights (1400 lumens), set to 20°C and given a 24hr light cycle to better facilitate growth. Each of the three batches was stored on a separate shelf of the chamber, with the placement of each replicate randomized. After ~8 weeks of growth, water was changed one final time. Then, ~3 days after

the last water change, replicates were removed from the environmental chamber and taken for use in a feeding trial.

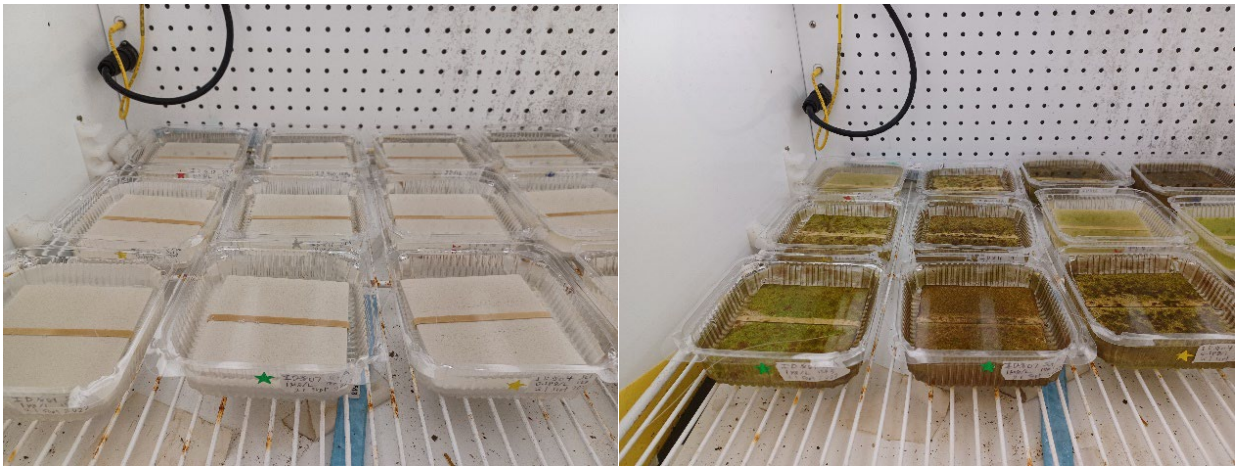

Figure 4: Photographs of tiles within the environmental chamber at the start (left) and end (right) of the growth period.

Feeding Trials

Prior to the feeding trials, snails were retrieved from their holding facility and temporarily housed in smaller tanks in the lab so as to control their food access, with different tanks designated for each batch of trials. Snails were fed regularly until 3 days prior to their feeding trial, after which they were not fed so as to incentivize feeding during the trial.

New plastic containers of similar dimensions and make were used for the feeding trial so as to ensure that the only periphyton present was on the tiles, not on the container walls (Figure 2, Step 3). These containers were filled with 250mL of stream water, with no added caffeine. This ensured that the only significant exposure the snails had to caffeine was coming from the periphyton rather than directly from the water, enabling a focus on indirect, bottom-up effects. In each feeding trial container, a specially fitted barrier created from a laminated cardstock cutout was added. These barriers fastened around the edges of the rubber band bisecting the tile and

restricted the snails (with a ~98% success rate) to only accessing half of the tile. This enabled me to preserve half of each tile ungrazed by the snails for my analysis (Figure 5).

Before commencing the feeding trials, carefully angled photographs were taken of each tile to enable a before/after comparison of the snails' feeding patterns (Figure 6). Once all barriers were in place and all "before" photographs had been taken, one snail was placed onto the middle of the "grazing" side of each tile and the timer was started on the feeding trial.

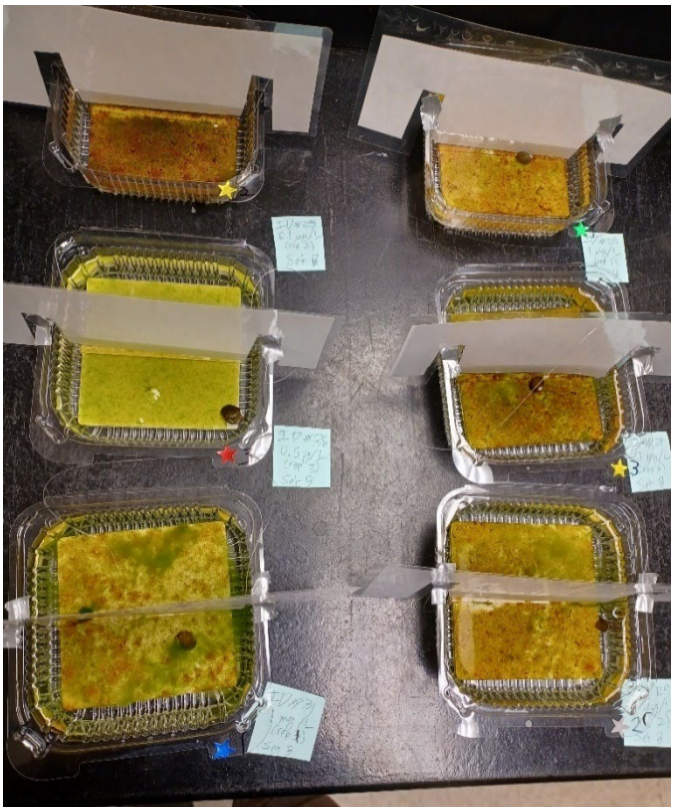

Figure 5: Photograph of feeding trial in progress, with snails on tiles bisected by laminated barriers.

The feeding trials lasted 4 hours. Every 10 minutes during the trial, three variables were recorded for each replicate: (1) the location of the snail, (2) whether it had visibly moved since the previous observation, and (3) whether it had left a visible grazing trail since the previous observation. At the end of the feeding trial, the snails were removed, placed into labeled

individual zip lock bags, and euthanized by freezing for later analysis. An “after” photograph was taken of each replicate tile using the same lighting and angle as in the “before” photograph.

Finally, the periphyton of each tile was scraped into three separate labeled vials. The entire grazed side of each tile was stored as one sample. The ungrazed side was divided into two equal areas and stored as two separate samples to enable different analyses. All samples were then frozen for later analysis.

Sample Measurements

Biomass estimates for the periphyton were determined by measuring the dry weight of the thawed samples of the grazed side and one of the two half samples of the ungrazed side, with these masses scaled proportionally to the area sampled. The ungrazed biomass would allow for direct assessment of the impacts of the caffeine on biofilm growth. By scaling them proportional to each other, I had hoped to directly measure reduction in periphyton biomass from grazing by subtracting grazed from ungrazed. However, this calculation included negative results, indicating that the biomass was too spatially variable for this approach to work. Consequently, I instead used the ungrazed biomass of known area (5cm x 4.7cm) to get a rough estimate of biomass density for each tile (mg/cm2), then used the grazed area measurements (cm2, below) to estimate grazed biomass (mg).

The area of periphyton grazed by each snail was assessed using Fiji, a pre-packaged version of ImageJ. Before and after photographs were paired for quick comparisons (Figure 6) and the scale of each image was set using the known dimensions of the tile. Areas of grazing were then visually identified on the “after” image, then selected using the “Wand” tool (8-connect mode, tolerance=10, no smoothing) so that their total area could be calculated (methodology broadly similar to McCluney & Sabo, 2009).

Figure 6: Example of paired photographs used for Fiji image analysis. The photograph on the left was taken before the feeding trial, while the photograph on the right was taken afterwards. Note the patch of exposed tile on the right image, which was measure using the "Wand" select tool in Fiji.

Additionally, a subset of eighteen tiles (one tile per treatment group per batch) were selected for nutrient analysis on a portion of the ungrazed periphyton biomass. Due to the earlier thawing process, some amount of cell lysis likely occurred, which may have impacted nutrient content of the dried biofilm. However, this process was consistent across all treatment groups. After the periphyton samples had been dried and weighed for the biomass estimates, they were ground up, weighed again, and submitted for analysis by a Discrete Analyzer in the Midden Lab at BGSU in order to determine the nitrogen and phosphorus concentrations (mg/kg) within each sample. Due to minimum mass requirements for the analysis, the highest biomass sample was used for each combination of batch and treatment group.

Data Analysis

Data sets on periphyton biomass, periphyton nutrient content, snail grazing quantity, and feeding trial behaviors were analyzed in R (version 4.3.1 "Beagle Scouts"). The periphyton's dry weight biomass (grazed and ungrazed) was scaled proportional to the known sample area. The phosphorus and nitrogen mass concentrations measured by the Midden Lab were combined with

my measurements of ungrazed dry biomass and its known sample area to calculate the total phosphorus and total nitrogen present within the whole tile of periphyton. This in turn was used to calculate the periphyton's N:P ratio. Measurements of grazed area were combined with ungrazed dry biomass and the known sample area to calculate an estimated grazed biomass. And for each replicate, the observations made during the feeding trial were combined into a time proportion of each response for location, movement, and grazing, with further composite proportions made to streamline analysis.

For each of these data sets, a one-way ANOVA was conducted comparing caffeine treatment groups, with an additional block to account for the three staggered batches that subdivided the 54 replicates. For the nutrient data, there were only 18 replicates (one per batch per treatment group), which somewhat limited the statistical power to detect differences between treatment groups across batches. Assumptions of equal variance and normality were assessed via plots of residuals, with transformations employed for situations in which tests did not satisfy assumptions. In particular, the grazed area and estimated grazed biomass were log transformed to account for a notable outlier, and all proportions were logit transformed (Warton et al., 2011).

RESULTS

Effects of Caffeine on Periphyton Biomass

Visual observations throughout the growth period suggested a pattern of delayed and reduced growth in the periphyton of the highest caffeine treatment group (0.5 g/L). This observation was also borne out in measurements of periphyton biomass, where tiles from the 0.5 g/L treatment group were observed to have significantly lower biomass than the other treatment groups ($F=7.384$, $df = [5, 46]$, $p < 0.0001$, Fig. 7), confirmed by a TukeyHSD test to be significantly different from all other groups ($p < 0.01$), none of which were significantly different from each other.

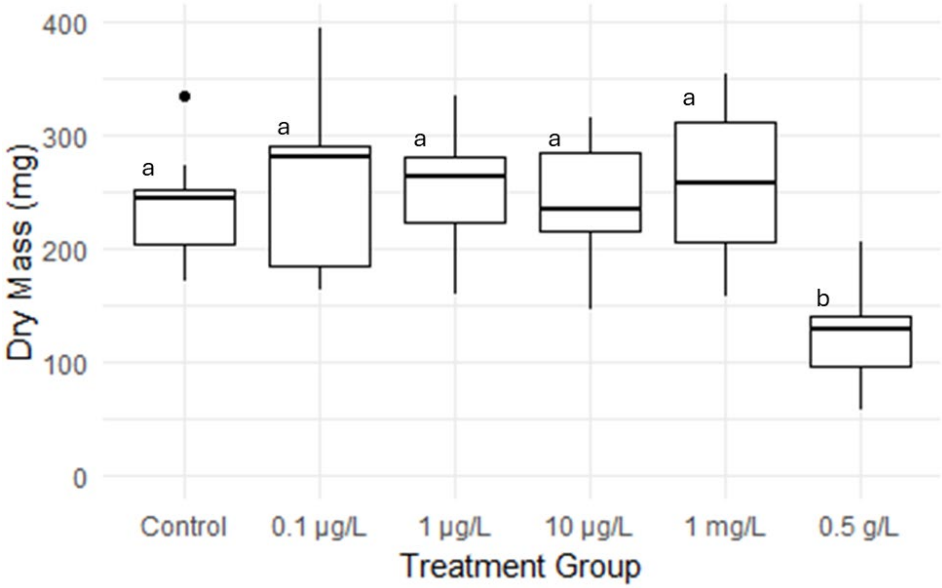

Figure 7: Dry weight biomass of periphyton across treatment groups, collected from the ungrazed side of the tile where snails were excluded. Only half of the ungrazed side (one quarter of the tile) was used for this mass measurement, which was then scaled to the size of the whole tile (quadrupled). Each of the six treatment groups had 9 replicates (N=54). The 0.5 g/L treatment group showed a significantly lower biomass than the other treatment groups ($p < 0.0001$).

Effects of Caffeine on Periphyton Nutrients

Measurements of the nutrient concentrations (mg/kg) in the samples analyzed revealed that the concentration of phosphorus was relatively consistent in most treatment groups, but

significantly higher in the 0.5 g/L treatment group ($F = 24.711$, df = [5, 10], $p < 0.0001$, Fig. 8), confirmed by TukeyHSD to be the only significantly different group ($p < 0.001$). By contrast, the concentration of nitrogen showed a gradual increase across treatment groups with increasing concentrations of caffeine ($F = 3.507$, df = [5, 10], $p = 0.0432$, Fig. 9). A TukeyHSD test found that the control group was significantly different from the 0.5 g/L treatment group ($p = 0.0451$), while none of the intermediate treatment groups were significantly different from either.

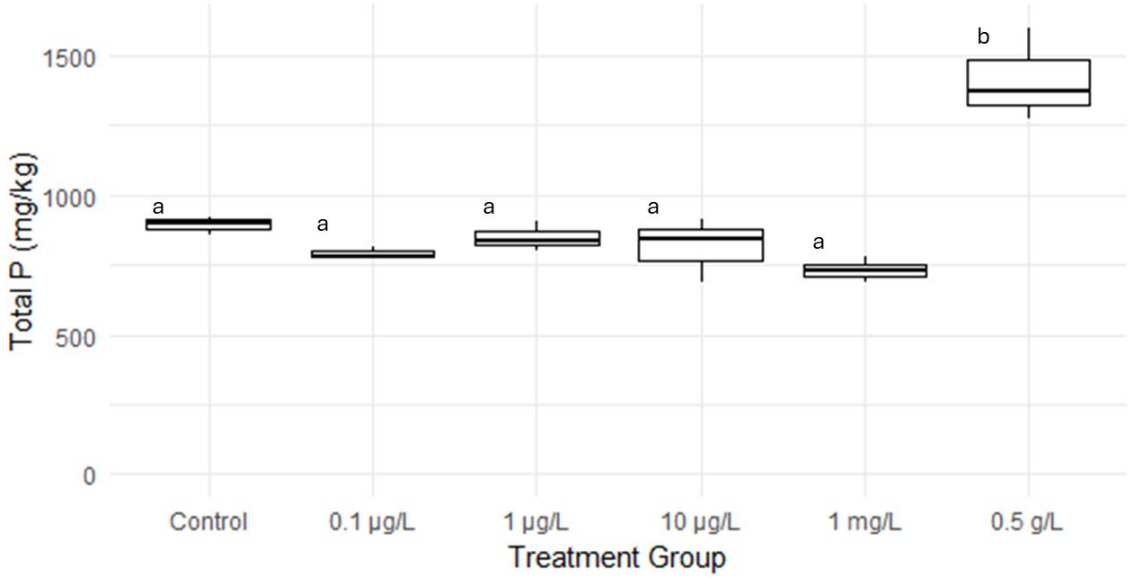

Figure 8: Concentration of phosphorus (mg/kg) in ungrazed biomass, measured in a subset of tile samples (N=18), with 3 replicates per treatment groups (one from each batch). The 0.5 g/L treatment group had a significantly higher concentration of phosphorus than the other treatment groups ($p < 0.0001$).

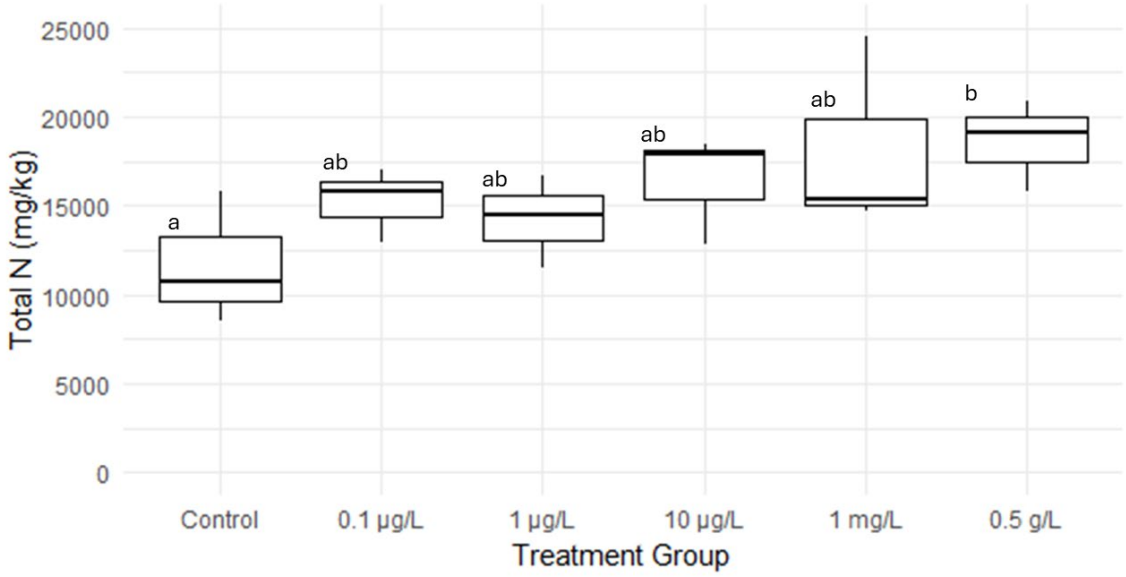

Figure 9: Concentration of nitrogen (mg/kg) in ungrazed biomass, measured in a subset of tile samples (N=18), with 3 replicates per treatment groups (one from each batch). The 0.5 g/L treatment group had a significantly higher concentration of nitrogen than the control group (p=0.0451), while the other treatment groups formed a more gradual transition.

By multiplying the concentration measurements by the ungrazed biomass measurements (scaled proportional to the whole tile), the total mass (mg) of each nutrient in the periphyton was calculated. With this adjustment, the total periphyton phosphorus proved to be fairly stable, without significant difference even in the 0.5g/L treatment group ($F = 1.470$, df = [5, 10], $p = 0.2821$, Fig. 10), which had higher concentrations, but lower biomass than other treatments. By contrast, the total periphyton nitrogen showed a more interesting pattern, with significantly elevated nitrogen levels in the middle concentration treatment groups ($F = 6.089$, df = [5, 10], $p = 0.0077$, Fig. 11). A TukeyHSD test showed that the 0.1 µg/L and 1 mg/L treatment groups were significantly different from both the control and 0.5 g/L treatment groups ($p < 0.05$), while the 1 µg/L and 10 µg/L treatment groups were not significantly different from either of the other sets.

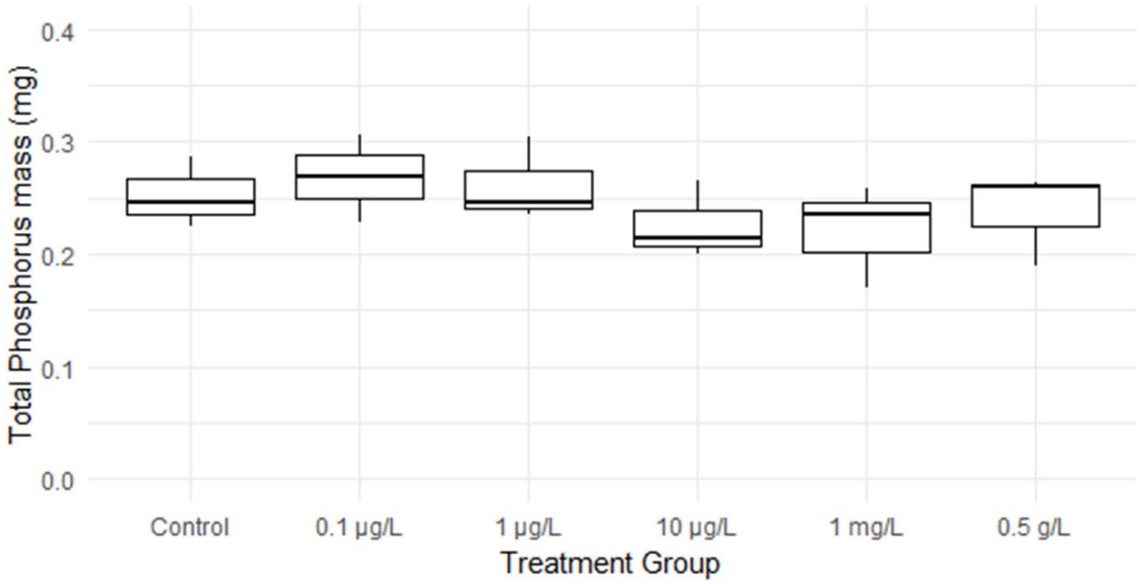

Figure 10: Total mass of phosphorus (mg) within the periphyton biofilm, scaled proportional to a whole tile, measured in a subset of tile samples (N=18), with 3 replicates per treatment groups (one from each batch). No significant differences were observed between treatment groups with regards to total phosphorus mass.

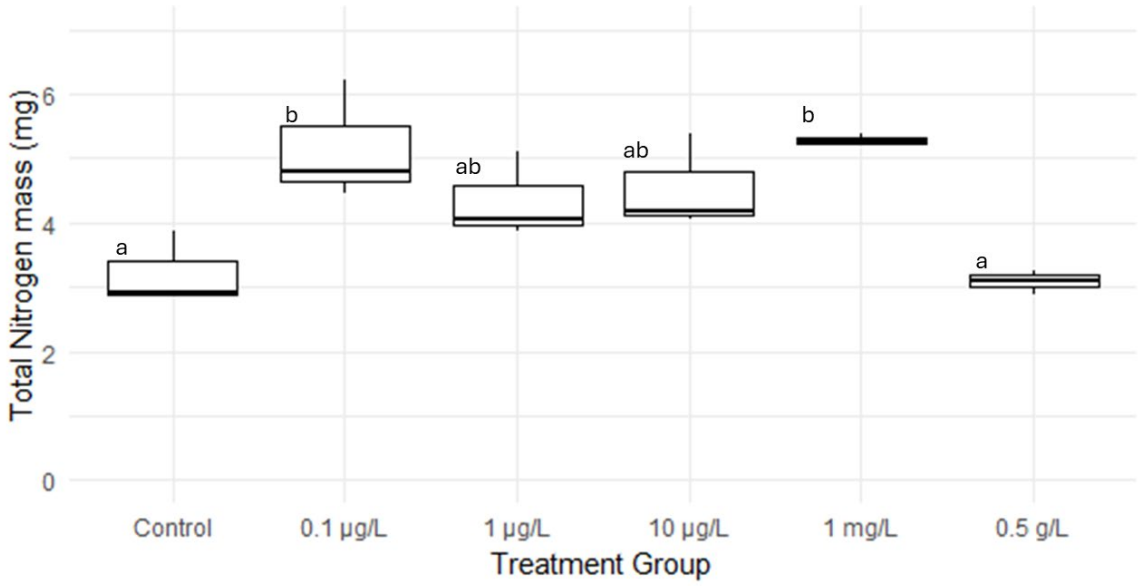

Figure 11: Total mass of nitrogen (mg) within the periphyton biofilm, scaled proportional to a whole tile, measured in a subset of tile samples (N=18), with 3 replicates per treatment groups (one from each batch). The 0.1 µg/L and 1 mg/L treatment groups had significantly higher nitrogen than the control and 0.5 g/L treatment groups ($p < 0.05$), while the other two groups were more intermediate.

Combining these two values to get the ratio of nitrogen to phosphorus showed a significant effect of caffeine that was broadly similar to the total nitrogen (F = 4.479, df = [5, 10], p = 0.0211, Fig. 12). The TukeyHSD test showed a significant difference between the 1mg/L treatment group and both the 0.5 g/L treatment group and the control group ($p < 0.05$), with other treatment groups (including 0.1 µg/L) not showing any significant differences.

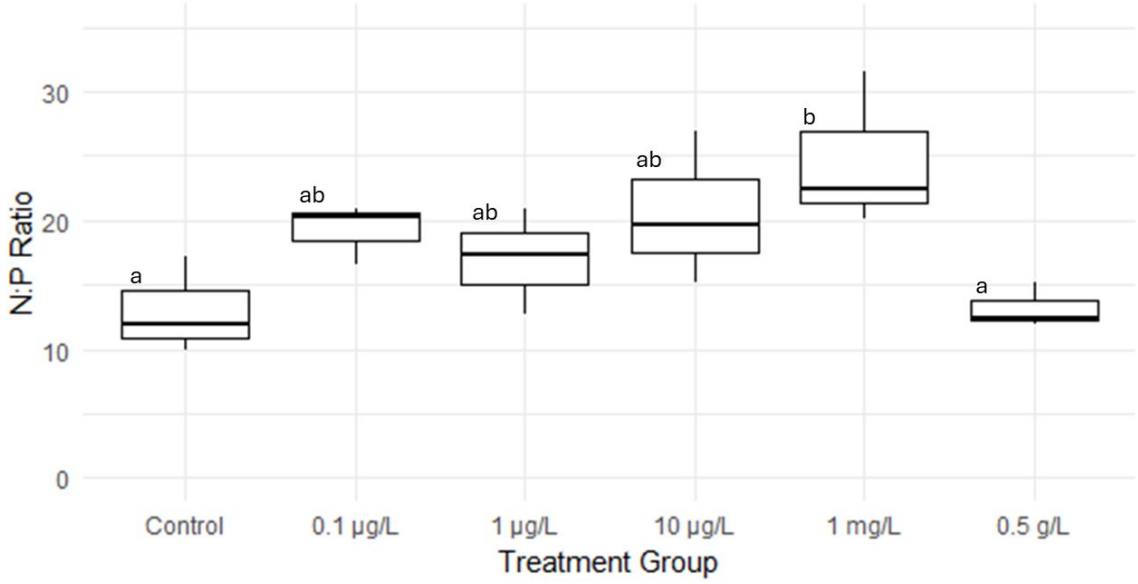

Figure 12: Nitrogen to phosphorus ratio within the periphyton biofilm, measured in a subset of tile samples (N=18), with 3 replicates per treatment groups (one from each batch). The 1 mg/L treatment had significantly higher N:P than the control and the 0.5 g/L treatment ($p < 0.05$), while the 0.1 µg/L, 1 µg/L, and 10 µg/L treatment groups were intermediate.

Effects of Caffeine-Treated Periphyton on Snail Grazing Outcomes

No statistically significant differences were observed between treatment groups with regards to grazing area (F = 1.189, df = [5,46], p = 0.329, Fig. 13). Most treatment groups showed substantial variance in the amount of grazing performed by the snails. There was a notable outlier in grazed area in the treatment group with the highest concentration of caffeine (0.5 g/L), where much more periphyton biomass was displaced than in any other replicate. To account for this, a log transformation was applied, though still with no significant differences detected (F = 1.279, df = [5, 46], p = 0.2893, Appendix Fig. 20).

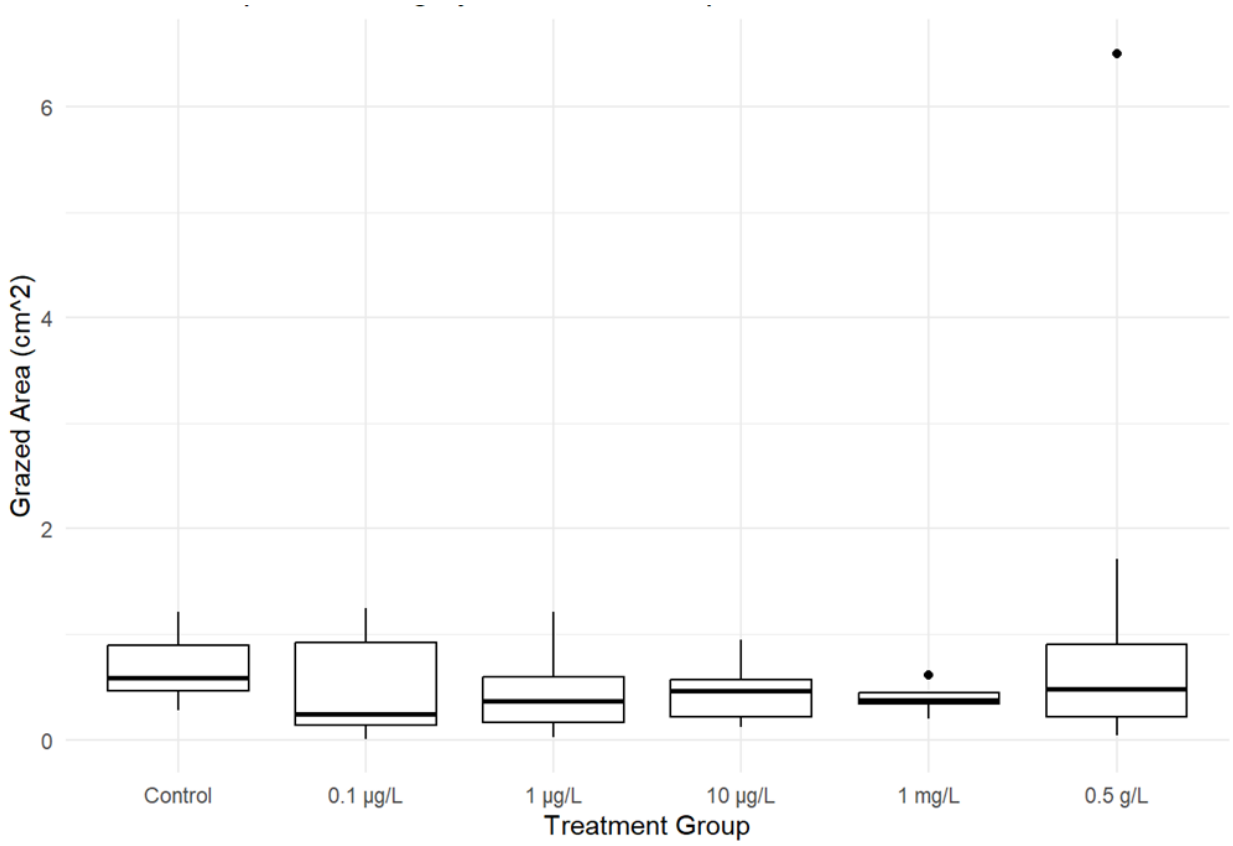

Figure 13: Area of complete grazing (cm2) measured in photographs of snail-grazed tiles using the wand tool in Fiji (ImageJ), with 9 replicates per treatment group (N=54). Despite a noticeable outlier in the 0.5 g/L treatment group, no significant differences were observed.

In order to estimate the grazed biomass, I calculated the average biomass density of the ungrazed side of each tile (using measured biomass and constant sampled area), then multiplied that by the measured area grazed for each tile. Due to previously mentioned variation in biomass between treatment groups, particularly the low biomass of the 0.5 g/L treatment group, the grazed biomass estimates showed a slightly different pattern than the grazed area measurements, though the notable outlier remained. Ultimately, there were still no significant differences found between treatment groups for grazed mass estimates ($F = 0.427$, df = [5, 46], $p = 0.8277$, Appendix Fig. 21), which remained the case with the application of a log transformation ($F = 1.093$, df = [5, 46], $p = 0.3773$).

Effects of Caffeine-Treated Periphyton on Snail Movement Behavior

Analysis of the snail behavior observations from the feeding trials also revealed some interesting patterns. For the location observations, I calculated the proportion of time points each snail spent in various locations within the trial container. Using these proportions to create a boxplot (Appendix Figure 18) suggested that snails in the 0.5 g/L treatment group were spending a reduced amount of time in the "Tile Center" and "Tile Edge" locations, with a similar but weaker pattern in the 0.1 µg/L treatment group. To better analyze this pattern, I calculated the combined proportions of time spent in "on tile" and "off tile" locations. Applying a logit transformation, I found a significant decrease in the proportional time spent "on tile" in the 0.5 g/L treatment group and, to a lesser extent, the 0.1 µg/L treatment group ($F = 9.601$, $df = [5, 46]$, $p < 0.0001$, Fig. 14). TukeyHSD testing showed that the 0.1µg/L treatment group was significantly different to the control and the 10 µg/L treatment groups ($p < 0.05$), while the 0.5 g/L treatment group was significantly different to every group but the 0.1 µg/L treatment group ($p < 0.001$).

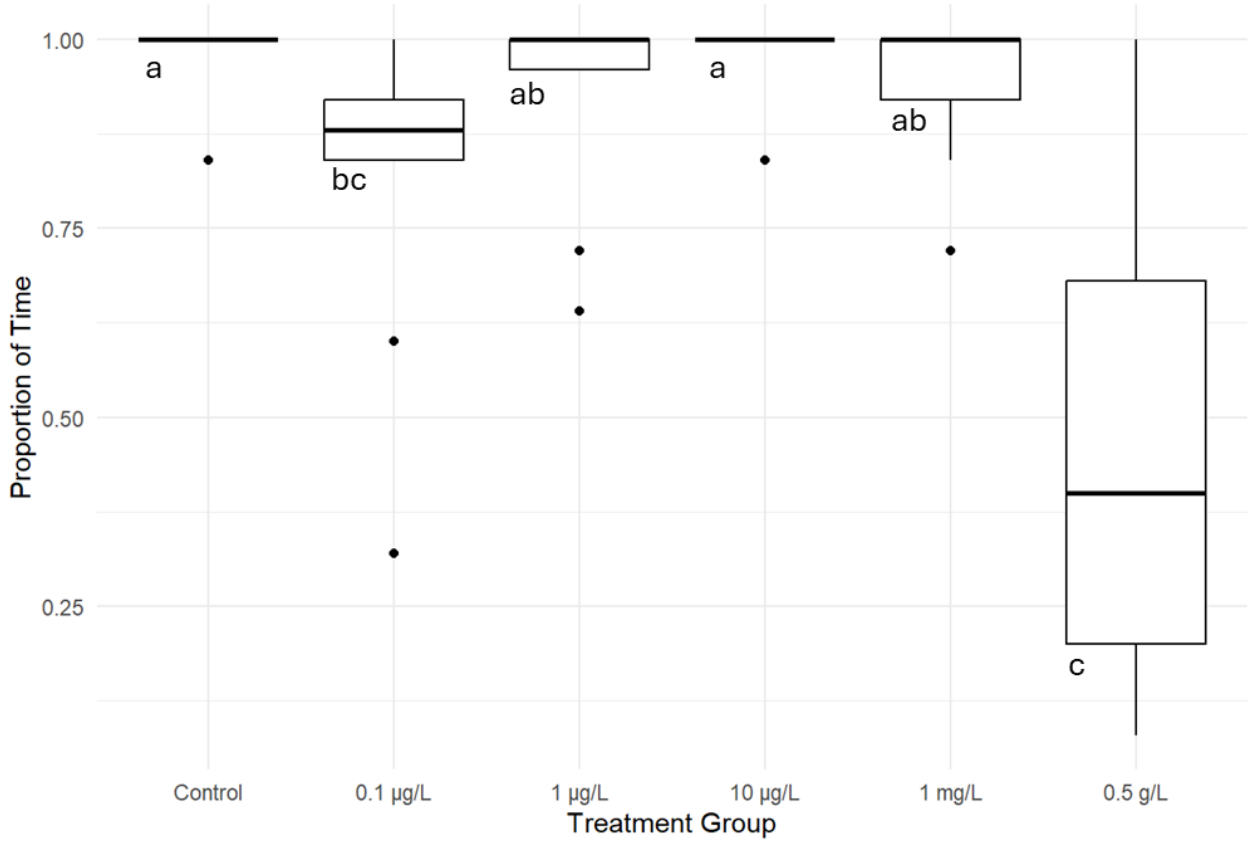

Figure 14: Proportion of time snails spent on the periphyton tile by treatment group, combining the "Tile Edge" and "Tile Center" observations for each of 9 replicates per treatment group (N=54). Snails in the 0.5 g/L treatment group spent much less time on the tile than other treatment groups except for the 0.1 µg/L treatment group ($p < 0.001$), which itself spent significantly less time on the tile than the control group or 10 µg/L treatment groups ($p < 0.05$).

For observations of active movement and active grazing by the snails, I calculated the proportion of time points with positive observations for each snail (visible shift and location and visible grazing trail, respectively). Applying a logit transformation, this movement data showed significantly increased movement levels in the 0.5 g/L treatment group ($F = 4.248$, df = [5, 46], $p = 0.0029$, Fig. 15). A TukeyHSD test showed that the 0.5 g/L treatment group was significantly different from every other group ($p < 0.05$) with the exception of the 0.1 µg/L treatment group (which had non-significant differences with other groups) and the 10 µg/L treatment group.

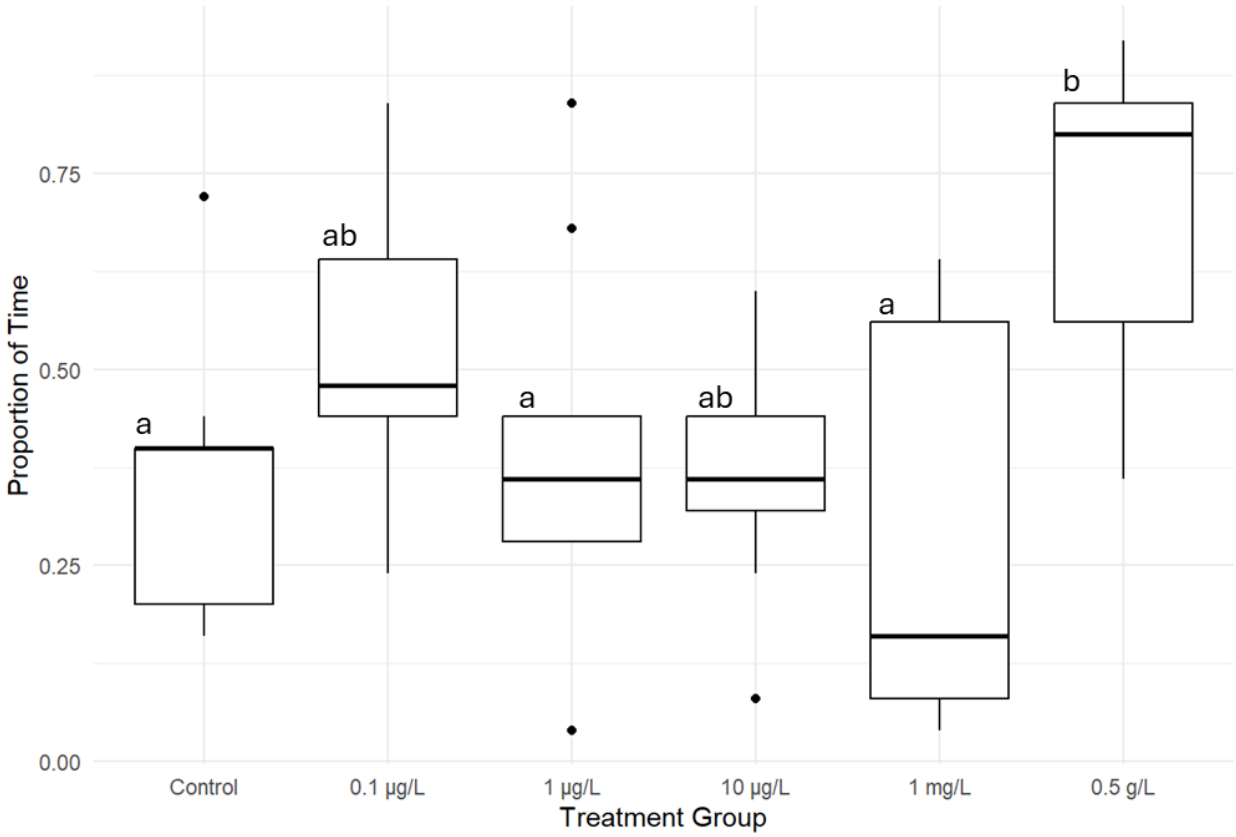

Figure 15: Proportion of time snails spent moving across treatment groups. Data is a composite of all "Y", "YY" (particularly noticeable movement), and "L" (noticeable but little movement) datapoints for each of 9 replicates per treatment group (N=54). The 0.5 g/L treatment group spent significantly more time moving compared to the control, 1 µg/L, and 1 mg/L treatment groups ($p < 0.05$).

Unlike location and movement, the observations of active grazing during the feeding trial did not show any statistically significant differences among treatment groups ($F = 0.336$, $df = [5, 46]$, $p = 0.888$, Fig. 16), which is consistent with the previously mentioned lack of significant differences in the measurements of grazed area and estimates of grazed biomass on the tiles.

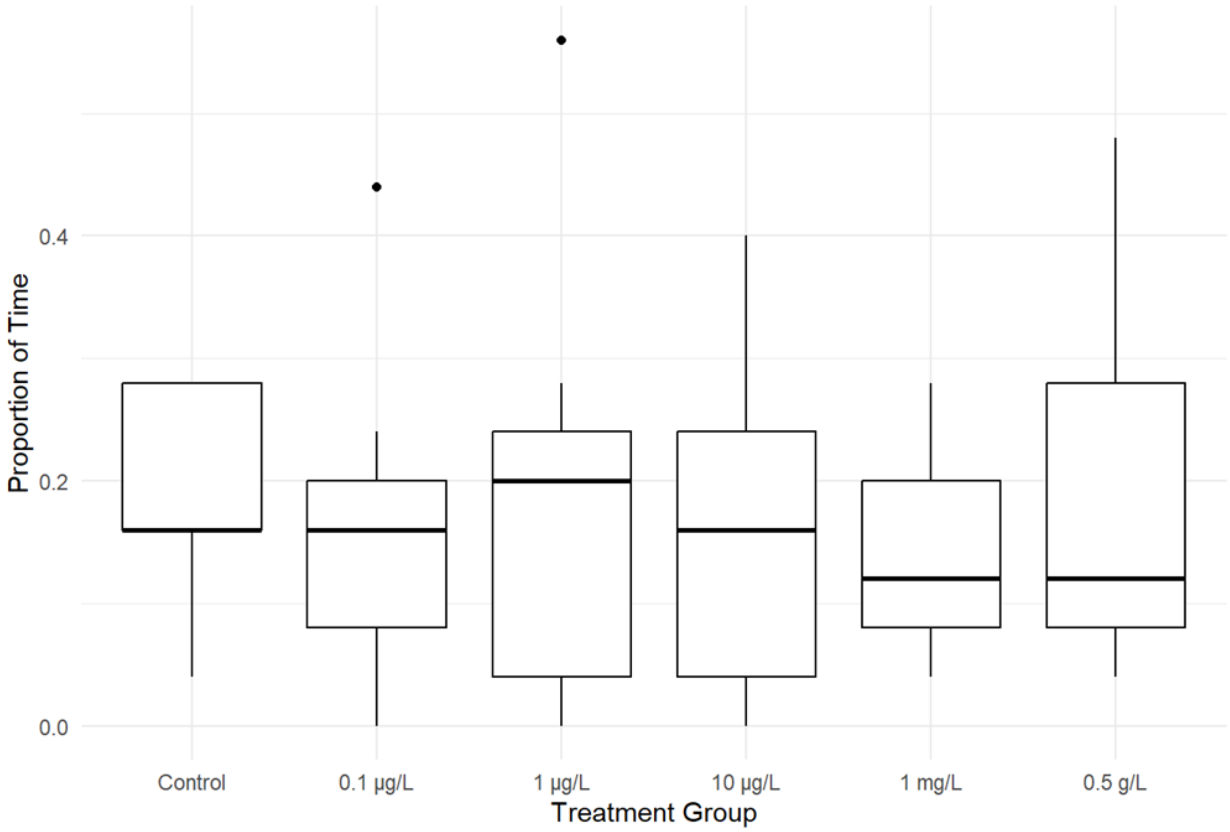

Figure 16: Proportion of time snails in each treatment group had new observable grazing on the tile. Data is a composite of all "Y" and "L" [little but noticeable grazing] observations for each of 9 replicates per treatment group (N=54). No significant differences were observed.

DISCUSSION

Exposure to caffeine had several significant effects, both directly on periphyton and indirectly on grazing snails. The most visibly obvious were observed in the 0.5 g/L concentration, namely the dramatic reduction in periphyton biomass and the significant increase in snails spending time off the periphyton tile. Both of these are consistent with a hypothesis of caffeine toxicity to periphyton at high concentrations. More broadly, at all lower caffeine concentrations the nitrogen content of the periphyton exhibited a significant increase alongside the increasing caffeine concentrations, suggesting an uptake of caffeine by the periphyton and potential use as a nitrogen source by some member of the biofilm (perhaps fungal, as speculated by Marshall et al., 2022, based on Hakil et al., 1999). However, the lack of significant increases to biomass suggests that something else is acting as a limiting factor and could indicate that caffeine was merely accumulating in the biofilm, not being utilized. Interestingly, a significant effect on snail grazing (whether area grazed, estimated grazed biomass, or observed frequency of grazing) was not detected. This may indicate that potential caffeine avoidance behavior is limited in scope (or perhaps has a delay after feeding) and may also suggest that the observed range of nutrient concentrations was insufficient to induce compensatory feeding or accelerated satiation, at least within this experimental system and time frame. It may also be a consequence of high variance in snail behavior, which additional replicates could help to resolve.

Impacts of Caffeine on Periphyton

The two broad patterns observed in the periphyton, altered nutrient content across concentrations and reduced biomass at high concentrations, are interrelated. The available phosphorus was not altered across treatment groups, and the generally consistent levels of phosphorus across most treatment groups line up with this. The spike in phosphorus

concentration in the 0.5 g/L treatment group corresponds with the reduced biomass of that treatment group: with roughly half the biomass, it had roughly double the concentration of phosphorus, and the total uptake of phosphorus within the periphyton did not change. This may suggest that phosphorus uptake is limited primarily by the amount of phosphorus available in the system, not the amount of periphyton doing the uptake, which might be the case if it is a limiting nutrient, or at least present at levels that do not exceed what the periphyton is capable of processing.

By contrast, the concentration of nitrogen per unit mass of periphyton was highest in the 0.5 g/L treatment, but total nitrogen uptake across the entire biofilm was significantly lower in this treatment compared to the 0.1 μg/L and 1 mg/L treatments, instead being very similar to the control group. This pattern is also attributable to the reduced biomass in the 0.5 g/L treatment: it may have had a concentration that was almost twice that of the control group, but it also had about half the biomass. Meanwhile, several more moderate caffeine treatments did show elevated nitrogen concentrations relative to the control, but since they retained a high biomass alongside this concentration, their total nitrogen was higher than both the control and the 0.5 g/L treatment. The fact that the 0.5 g/L treatment periphyton's nitrogen concentration did not further increase due to lower biomass demonstrates that the periphyton was not taking up all available nitrogen as was likely the case with phosphorus, which indicates that nitrogen was not a limiting nutrient.

While nitrogen showed a significant pattern across several treatment groups, both nitrogen and phosphorus and the overall N:P ratio were significantly impacted by the drop in biomass in the 0.5 g/L treatment group. It seems most likely that this drop in biomass was induced by the toxicity of caffeine at these high concentrations, either killing or inhibiting the growth of significant portions of the periphyton community. Anecdotally, I also observed that, in

addition to a lower biomass at the end of the experiment, the 0.5 g/L treatment significantly lagged behind the other treatments in terms of its growth. While my data show no clear evidence of intermediate toxicity effects at lower concentrations than 0.5 g/L, there is a wide range of untested concentrations between 0.5 g/L and 1 mg/L in which these effects might start to appear. It may also be that there were toxic effects at lower concentrations, but that they were balanced out by other factors. For instance, if a certain dose of caffeine is toxic to some taxa but a nitrogen source to others, the net impact on the periphyton biomass may balance out in contexts where nitrogen is a limiting factor.

My data do not enable elaboration on whether the increased nitrogen in the periphyton is found as normal periphyton biomass, perhaps with a flexible stoichiometry (Marshall et al., 2022), or if it is still caffeine that is simply accumulating within the biofilm without being utilized. Additional testing, such as measuring the concentration of caffeine directly within a periphyton sample using a mass spectrometer (de Sousa et al., 2021), could potentially help to resolve this, which might inform the likelihood of both caffeine utilization and of secondary caffeine exposure in grazers.

Relatedly, performing genetic analyses such as 16s/18s amplicon sequencing could provide information as to the community composition of the periphyton, which might provide insight as to which taxa are affected by caffeine toxicity at which concentrations and which taxa if any can utilize caffeine as a nitrogen source. Anecdotally, the periphyton on tiles displayed a diversity of colors and visual patterns, potentially suggesting a diversity in community structure both between tiles and in terms of spatial heterogeneity. And more notably, while processing the periphyton samples, I observed that the 0.5 g/L treatment group had markedly different color than other treatments, with more bright greens and even blues compared to the drabber, brownish

greens with hints of purple in the other treatments. Similarly, the 0.5 g/L treatment periphyton had a distinct texture, considerably looser and less coherent than the coarser biofilm in the other treatment groups. These suggest significant differences in at least the physical structure of the community, which likely imply differences in its taxonomic composition.

On a broader ecological level, my data only show reductions in periphyton biomass from caffeine at concentrations higher than the highest field observations I could find in the literature (1 mg/L, from Spongberg et al., 2011). However, there are important caveats to this. I only measured biomass at a single time point at the end of 8 weeks, potentially missing differences earlier in the periphyton growth cycle. My experimental design also limited certain temporal variances that exist outside the lab, such as daily shifts in light and temperature, repeated exposure to grazing pressures, as well as daily and weekly fluctuations in the anthropogenic inputs of contaminants like caffeine (McCluney Lab, Unpublished Data). I was also considering caffeine in isolation of the complex suite of co-occurring wastewater pharmaceuticals with which it is typically found (Marshall & McCluney, 2021). When factors such as these are taken into account, it is possible that the specific concentration at which caffeine starts to directly impede periphyton biomass could be different than what my data suggests.

Furthermore, the effects of caffeine on periphyton nutrient content were seen at much lower concentrations than the effect on biomass, with some effects seen even as low as the 0.1 µg/L treatment comparable to that in many streams. The degree to which this has an impact may depend on whether the caffeine is being utilized or simply bioaccumulating. If it is being utilized, then its impact may depend on the availability of other nitrogen sources. WWTPs tend to be significant point sources of nitrogen, with total nitrogen in effluent often in the range of 5 to 20 mg/L (Zhou et al., 2022; Wu et al., 2019). Calculating from the molar mass of caffeine

(194.194 g/mol), its chemical formula (C8H10N4O2), and the atomic weight of nitrogen (14.007), I find that my 0.5 g/L treatment significantly exceeds this concentration at 144.25 mg N/L, while my 1 mg/L treatment has a more modest mass of 0.2885 mg N/L, with the other treatments orders of magnitude lower. As such, many of the most common sources of anthropogenic caffeine may co-occur with enough other anthropogenic nitrogen sources to make the input from caffeine negligible. However, the widespread usage of caffeine compared to other pharmaceutically active compounds increases the likelihood of it being introduced in relatively undeveloped areas with lower anthropogenic nitrogen inputs. Furthermore, facilities dedicated to the production of caffeine products may introduce additional significant point sources of caffeine specifically, as seen in Spongberg et al. 2011.

Beyond impacts on biomass and nutrient content, the impacts of caffeine on the community composition of periphyton remain somewhat understudied. Similarly, while this lab study emphasized bottom-up ecological effects, in real ecosystems these would interact with top-down effects in continuous feedback. As such, understanding the impact of caffeine exposure, direct and indirect, on other members of the ecosystem is also important.

Impacts of Caffeine-Exposed Periphyton on Snails

While this experiment measured direct effects of caffeine on periphyton, it was designed to only measure indirect effects on snails as grazers of periphyton. This is in deliberate contrast to typical ecotoxicological studies, which examine the direct impacts of contaminants on various organisms. While in a natural environment, both factors would be in play and interacting with each other, studying these indirect effects experimentally can help to develop a more holistic understanding of how each factor contributes to the overall effect.

No significant difference in the area of grazing, the proportion of time spent grazing, or the estimated mass grazed was observed across the six treatment groups from the four-hour feeding trial of this experiment. This may suggest that the immediate indirect effects of caffeine on snail foraging behavior via periphyton are limited. Longer-term effects of altered periphyton on health, growth, and reproduction need to be investigated.

Despite the lack of effect on foraging, there were some notable changes in other specific snail behaviors, most notably in the increased mobility and time spent off of the periphyton tile in the 0.5 g/L treatment group (with a similar but weaker effect also observed in the 0.1 µg/L treatment group). Several specific causal relationships that might exist. First, if there is residual caffeine present sequestered within the biofilm, this may induce an avoidance behavior in the snails, as caffeine has been shown to be toxic to snails in some contexts (Wszelaki et al., 2023) and has specifically been investigated as a potential snail deterrent (Horgan et al., 2023; Hollingsworth et al., 2003).

Secondly, it might also be possible that residual caffeine present in the biofilm could have a direct pharmaceutical effect on the behavior of snails that ingest the biofilm. At certain concentrations, caffeine has been observed to increase activity on some invertebrates as it has on many vertebrates (Mustard, 2014). If this is the case, then increased movement of the snails within the limited confines of the feeding trial container might cause them to spend more time off of the periphyton tile simply as a natural consequence of increased mobility.

It is plausible that caffeine was present within the periphyton in the 0.5 g/L treatment, since even if the biofilm does break down caffeine, the high concentration present and the lower biofilm biomass may have resulted in more residual caffeine left over. However, the behavioral effects observed at the 0.1 µg/L treatment are harder to explain.

If residual caffeine was not involved in the behavioral changes, there could have been a change in the community composition, altering the abundance of preferred or avoided taxa. It may also be possible that the reduced biomass of the periphyton in the 0.5 g/L treatment was perceived as a low-quality food source and that the snails were motivated to explore for better sources.

There were also a few interesting outliers that may suggest potential interactions between caffeine, periphyton, and snails. The largest single area of grazing measured by a significant margin was found in the 0.5 g/L treatment group. However, much of the loss of periphyton area on this tile seems related to dislodging the biofilm rather than strictly eating it, with noticeable quantities of periphyton drifting loosely in the container. Given the reduced biomass present in this treatment group and potential changes in taxa indicated by its change in color and texture, it seems plausible that the biofilm was less cohesive and more easily fragmented than in other treatment groups. This might have broader ecological implications for cases in which caffeine or other contaminants impede the growth of periphyton. It seems plausible that in a stream setting, especially one with variable flow such as is found downstream of many WWTPs, even changes in water velocity might threaten to dislodge such a low-integrity biofilm or at least impede its growth.

Similarly, one snail out of 54 managed to breach past the barrier I used to exclude the snails from the “ungrazed” half of the periphyton tile on multiple occasions during its feeding trial. While this may be simply indicative of a flaw in that particular barrier, allowing the snail space to squeeze past, it is interesting to note that this snail was feeding on the 0.5 g/L treatment periphyton. If lingering caffeine within this treatment is inducing increased movement in the

snails, this may serve as a striking example of how human pharmaceuticals can inadvertently affect the behavior of animals.

In a real-world ecological context, direct exposure to both caffeine and other co-occurring contaminants in the water column (Marshall et al., 2022) will frequently eclipse the indirect effects and secondary exposure I have examined in this study. However, while concentrations of caffeine and other pharmaceuticals in the water column are prone to fluctuations over time from pulses in the WWTP point sources (McCluney Lab, Unpublished Data), indirect effects such as bottom-up influences on periphyton may be more consistent. While the fluctuations in exposure these pulses cause have the potential for complex effects of their own, understanding the underlying persistent baseline changes that they leave behind is also important for developing a holistic understanding of how these pharmaceuticals and all anthropogenic effects alter aquatic ecosystems.

CONCLUSION

The numerous pathways by which pharmaceuticals such as caffeine and other contaminants of emergent concern can affect aquatic ecosystems can present a challenge for researchers looking to assess their total impact on an ecosystem. While field observations can assess the net impact of a particular contaminant on a given ecosystem, it can be challenging to isolate the specific impact of the contaminant from the countless other ecological factors present in the system. And while laboratory observations can help to assess the direct toxicological effects of particular contaminants on various organisms in isolation, they can lack the contextual information to assess indirect ecological effects. My experiment has attempted to explore some of these indirect ecotoxicological interactions to gain a better understanding of how they operate while still maintaining the benefits of a controlled laboratory setting. My results demonstrate that caffeine specifically can have a direct effect on periphyton nitrogen content even at low concentrations, and that at high concentrations (fortunately above most environmentally realistic levels) it can significantly limit periphyton growth and biomass. I have also demonstrated that, while direct measurements of snail grazing show minimal differences across periphyton treated with different caffeine concentrations, certain other behavioral traits do vary significantly in response to caffeine in this more indirect way, such as general movement level and time spent in proximity to caffeine-treated periphyton. While further testing and methodological refinements would be necessary to better quantify the various specific interactions between caffeine, periphyton, and grazing snails, the basic methodological framework developed could prove useful in exploring a wide variety of indirect ecotoxicological interactions.

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APPENDIX A. ADDITIONAL FIGURES

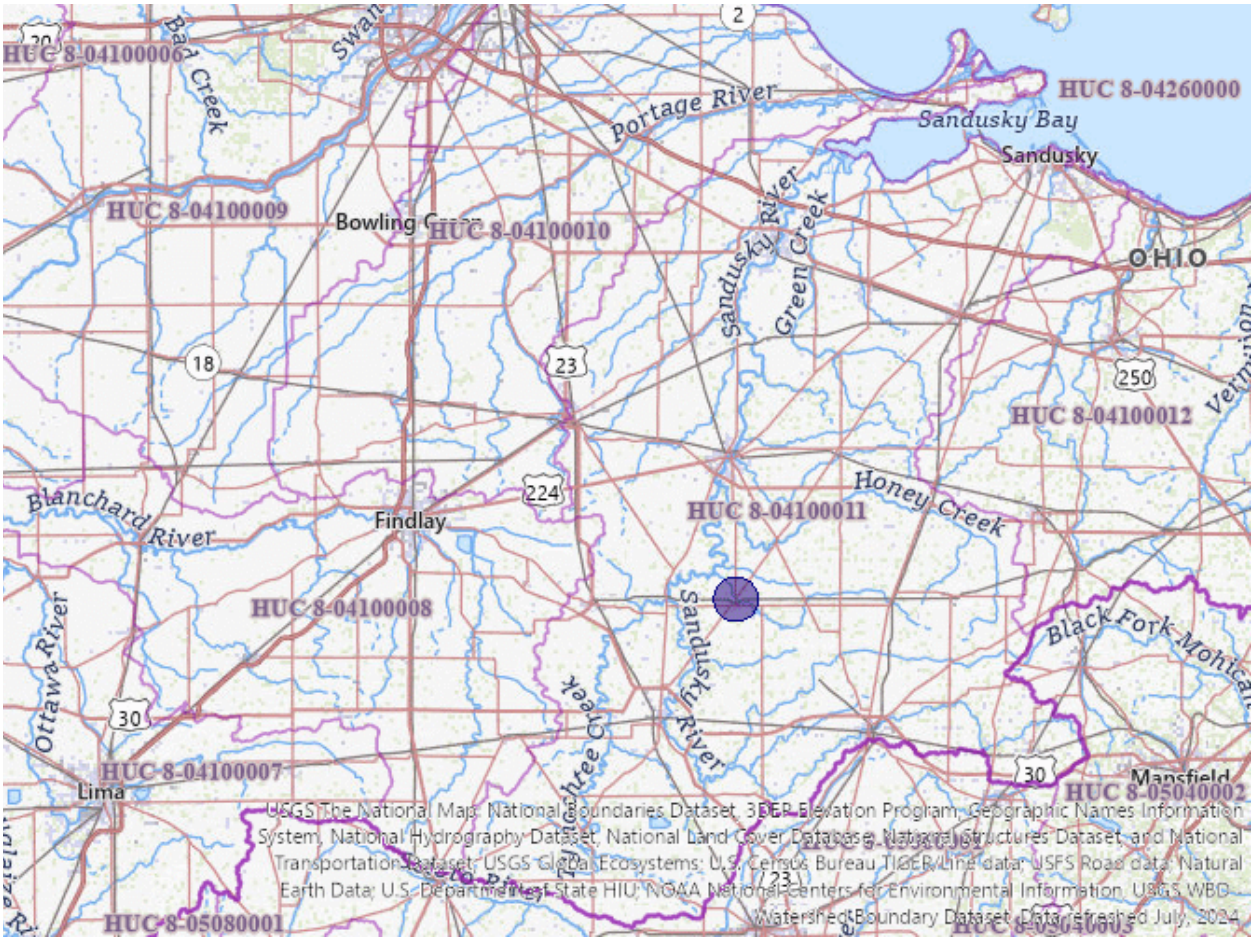

Figure A.1: Watershed map from the USGS showing the location of Sycamore Creek where water and periphyton samples were collected.

https://www.usgs.gov/faqs/where-can-i-get-watershed-maps

https://apps.nationalmap.gov/viewer/

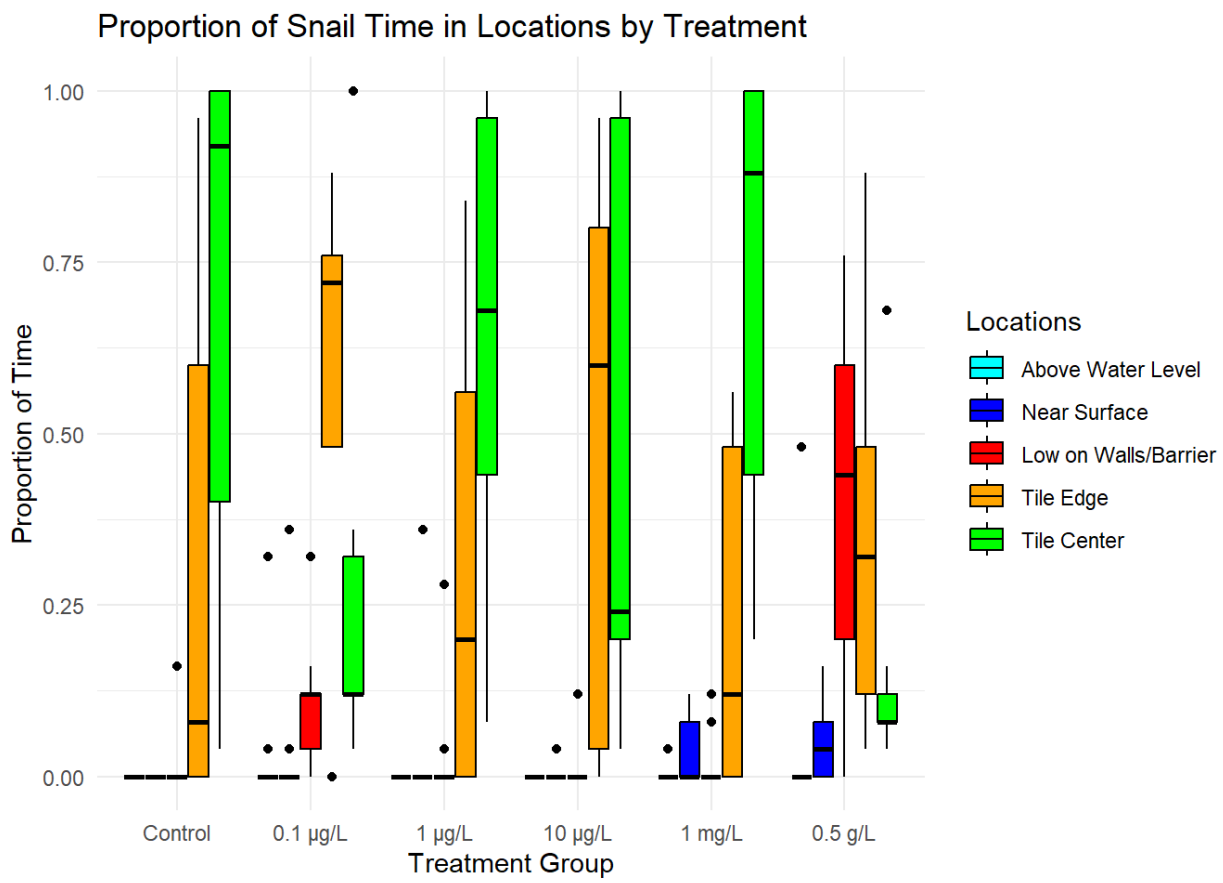

Figure A.2: Proportion of time snails spent in each of several locations within the feeding trial arenas. All snails were initially placed on the center of the grazing side of each tile, and their location was observed every 10 minutes.

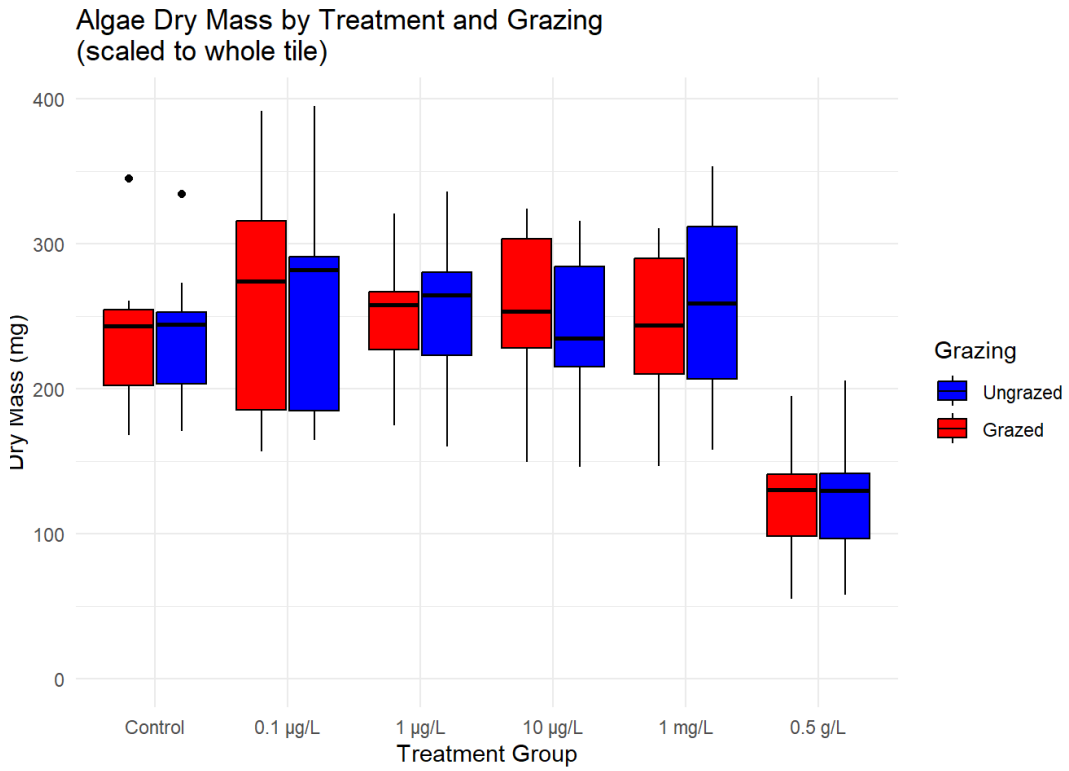

Figure A.3: original periphyton biomass graph, additionally contrasting grazed and ungrazed periphyton.

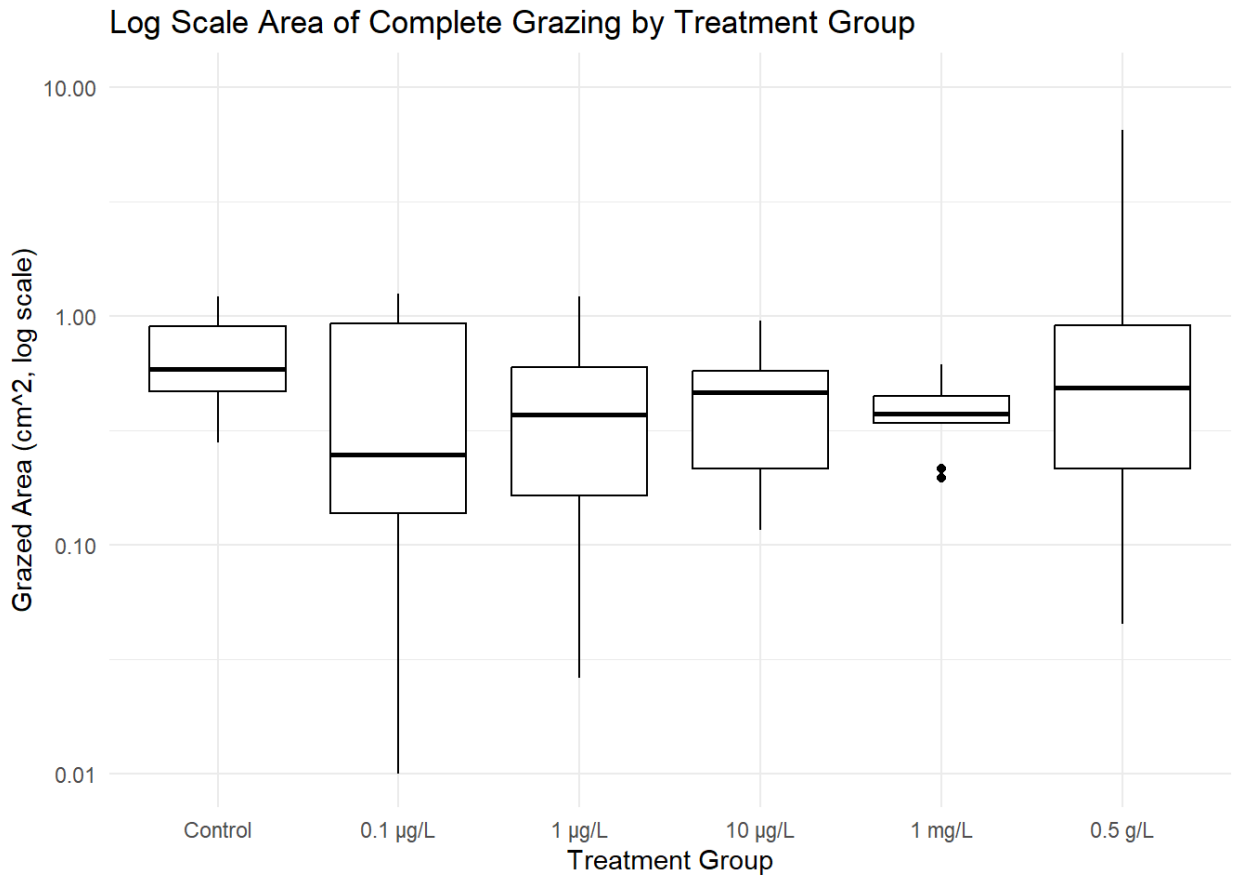

Figure A.4: Log scale version of Figure 13.

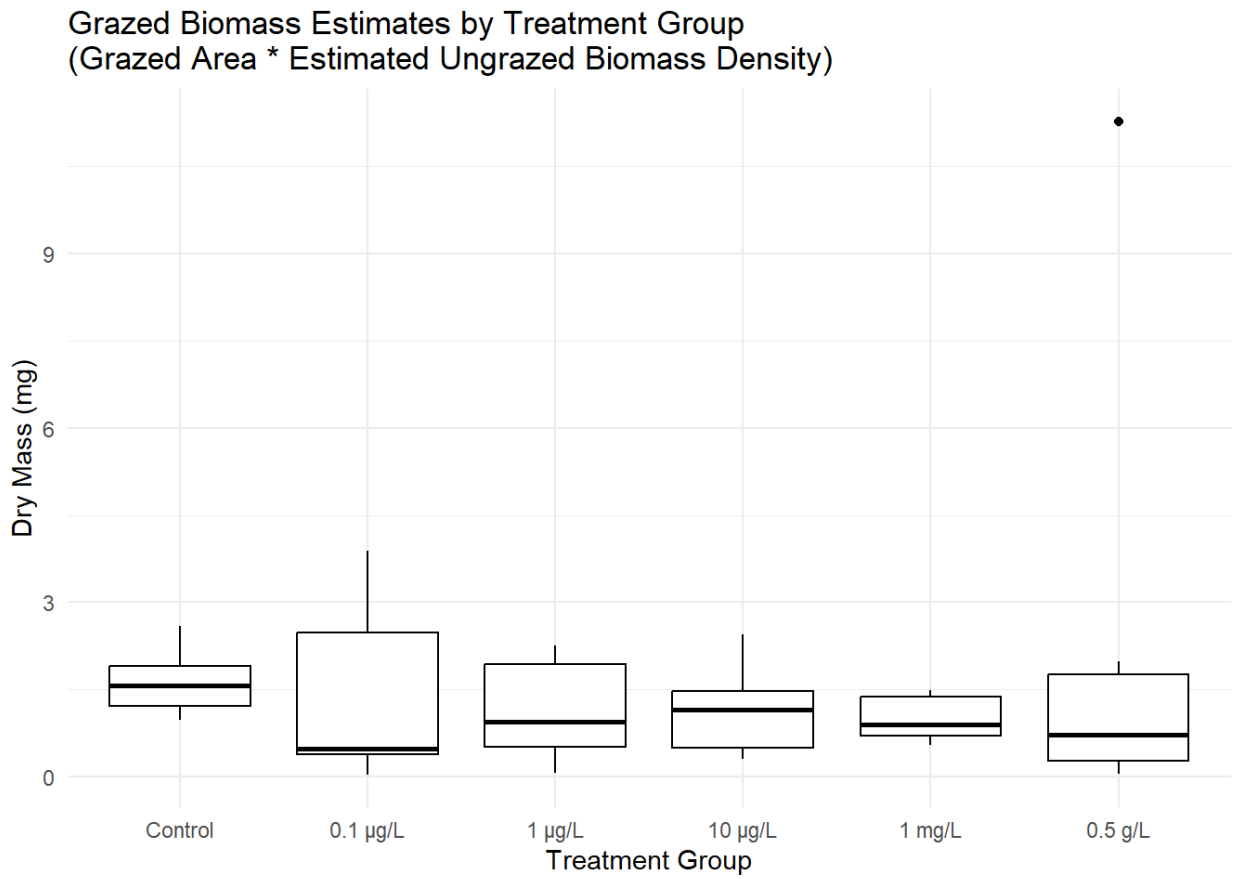

Figure A.5: Estimate of grazed biomass (mg), from grazed area and calculated biomass density.

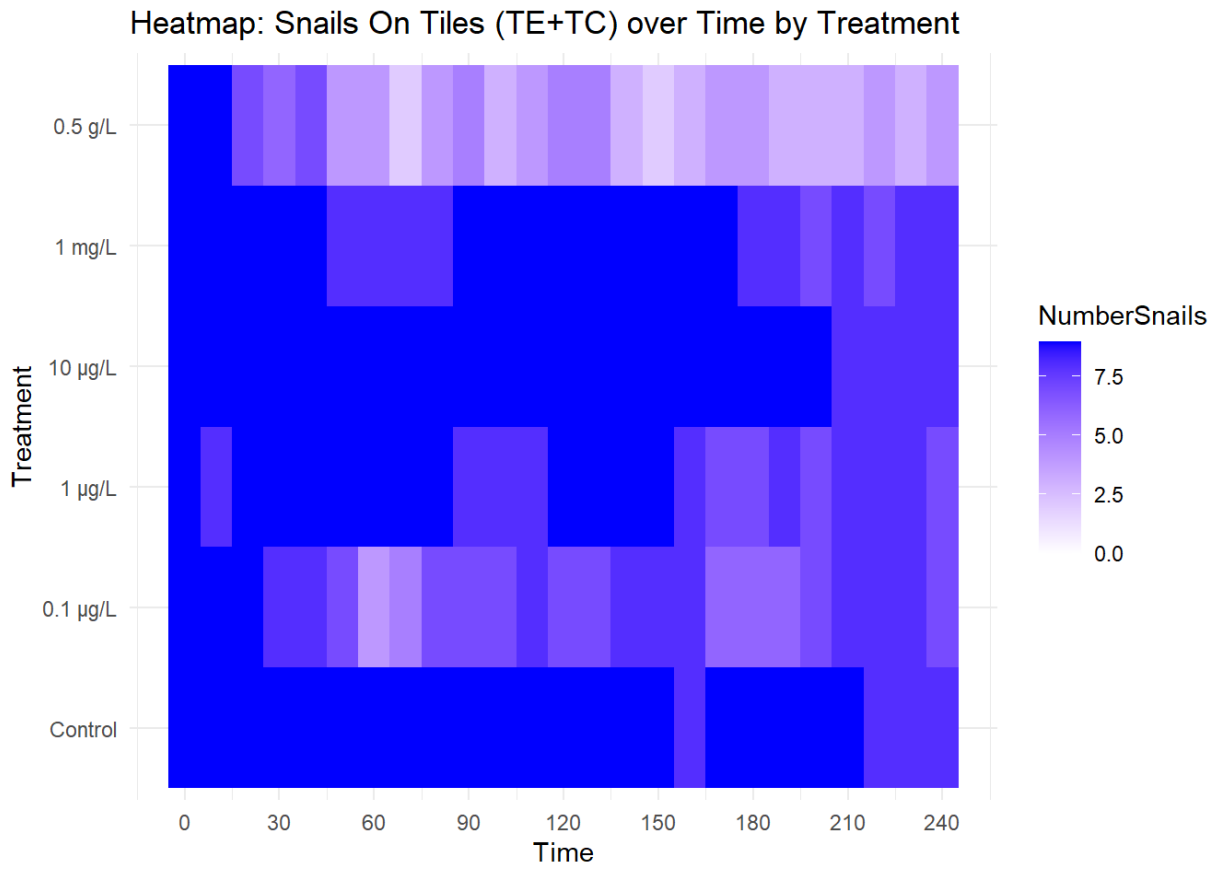

Figure A.6: Heatmap of snail locations over time by treatment group. All snails that are either on the Tile Center or Tile Edge are counted for each of the 25 time points observed.

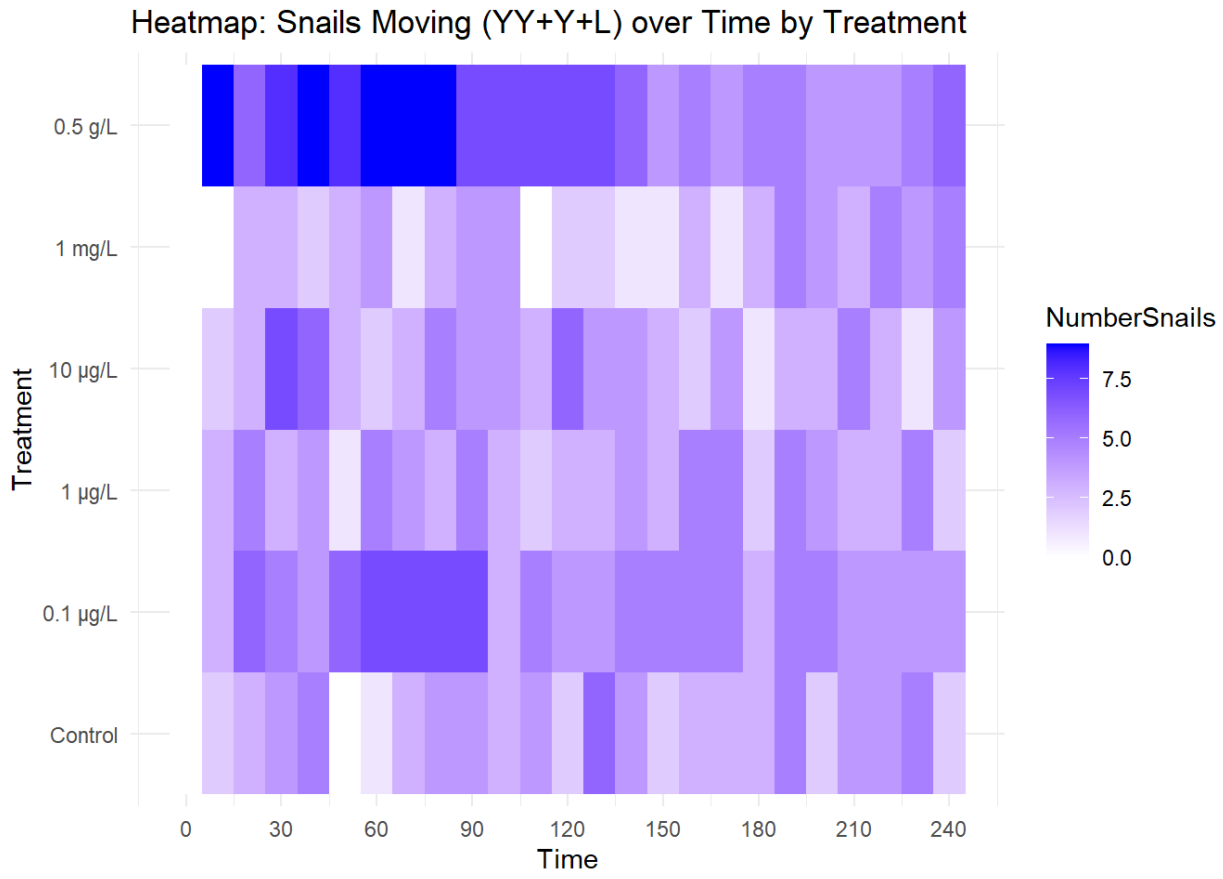

Figure A.7: Heatmap of snail movement over time. For each treatment group and time point, the number of snails that had noticeably moved ("Y", "YY", or "L" observations) is added up.

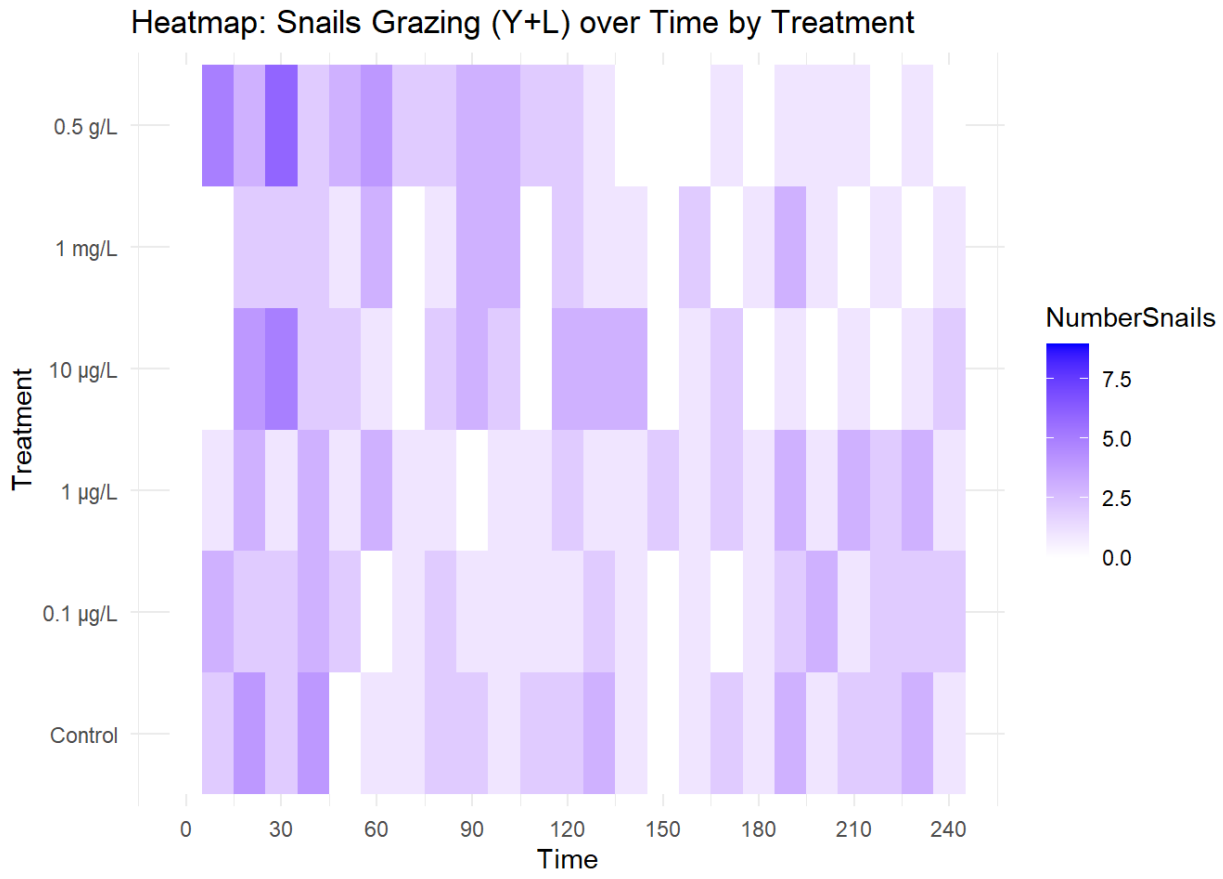

Figure A.8: Heatmap of snail grazing behavior for each treatment group over time. At each time point, the total number of snails who had visibly grazed since the previous time point is added up for each treatment group.